

Square Values

Teacher's Edition

Standard and Targets

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Boundary Square roots of perfect squares up to 225. Cube roots of perfect cubes from -125 to 125 . Students are not required to isolate the x^2 or x^3 term.

Learning targets I can square a number. I can find the side of a square from its area. I can explain why the graph of $y = x^2$ never goes below the x -axis.

Essential questions What does it mean to square a number? What is a perfect square? How do the geometry and the algebra of squaring line up?

Building on MA.6.NSO.3.3 and MA.7.NSO.1.1 — whole-number exponents with rational bases — and MA.6.GR.2.x for the area of a rectangle. Math Nation lists “N/A” here, which is not right: students have squared numbers before, they have simply never been asked to reverse the operation.

Working toward MA.8.NSO.1.1 in Lesson 4, where the side of a square with area 8 gets a name, and MA.8.GR.1.1 in Unit 6, where the Pythagorean Theorem asks for a square root on every problem.

Where this sits in the course MA.8.AR.2.3 is a Grade 8 FAST benchmark, reported under **Algebraic Reasoning**, which was the lowest-scoring category on the 2025 released items. It is also one of the few benchmarks where a single released item is public: a select-all question on the solutions of $x^3 = 64$ was answered correctly by **26% of students** — among the six lowest on the whole exam, tied for fifth. That item is Lesson 3’s work, and this lesson is what makes it possible.

Materials Whiteboards and markers. Calculators are allowed but should not be needed — every value in this lesson is on the perfect-square list. Worksheet 7A.02.01. Consider posting the list of perfect squares to 225 as a wall anchor chart for the whole unit.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same and different	2	4 min
Notes I–IV: squaring, the perfect squares to 225, area runs both ways, never negative	3–6	12 min
Example 1: areas, then backwards to the side	7–12	6 min
Example 2: the table and the picture	13–15	5 min
Example 3: no whole number squares to 8 — and no fraction either	16–18	5 min
Whiteboards: 9 questions, each with its reveal	19–36	15 min

Segment	Slides	Time
IXL — start today's skills	37	5 min
Total		53 min

Where the lesson lives: Notes IV (slide 6) and Example 3 (slides 16–18). Notes IV carries the one careful sentence about the origin that the textbook version gets wrong. Example 3 is the question that makes Lesson 4 inevitable.

Read this before you plan the day. Math Nation's own component times total 45 to 50 minutes before any worksheet, and the whole lesson is built as two student-directed partner Explorations. The plan above fits 53 because both Explorations are taught from the front as Notes II and III with Examples 1 and 2 as the worked versions. Students still meet every idea; they meet it as notes and independent practice rather than as discovery in pairs.

No partner or group work anywhere in this lesson. Math Nation's version has eight printed instances plus a Stronger and Clearer Each Time routine. Two of them needed real rebuilding rather than rewording: “form a group with another pair of partners” in the Exploration, and the definition table in Bring It Together where students collect two other pairs' definitions and have them initial the page. That signature table is now Notes I's vocabulary band, written once and copied.

Vocabulary

square (a number) — to multiply the number by itself

perfect square — a number you get by squaring a whole number

area — the number of unit squares that cover a figure

side length — the distance along one edge, always positive

A note on the glossary. Math Nation lists no key terms and no additional terms for this lesson, which is odd for a lesson whose entire point is a definition. The four above are the ones the worksheet uses. “Radical,” “radicand” and “index” arrive tomorrow and are deliberately not previewed here.

Slide-by-Slide Teaching Notes

Slide 1. Square Values *1 min*

- Post the learning target.
- Unit 1 was mostly grade 7 content these students had not met; Unit 2 opens on MA.8.AR.2.3, which the Grade 8 FAST reports under Algebraic Reasoning — the lowest-scoring category on the 2025 released items.
- Today is the concrete day.
- Tomorrow is the notation, and the geometry is what will make tomorrow's symbol mean something, so do not rush it.

Slide 2. Warm-Up *4 min*

- TYPE: Same and different.
- INDEPENDENT.

- Expect: same — both are a number multiplied by itself, both answers are positive, both bases are one digit.
- Different — the sign of the base, and the values (49 against 25).
- Do not accept 'one is negative' without 'and it still comes out positive'.
- That observation is Notes IV, arrived at before it is taught, which is what this warm-up type is for.
- This replaces Math Nation's growth-mindset reflection, which contained no mathematics.
- OFF THE SLIDE, YOURS TO SAY: Both are squarings. One base is positive and one is negative — and both answers are positive.

Slide 3. Notes: I. Squaring means multiplying a number by itself 3 min

- Students copy all three lines.
- Item C is the misconception Florida's own instructional guide names FIRST for this benchmark: students believe squaring means multiplying by 2.
- Write $5^2 = 10$ on the board, ask what is wrong with it, then correct it.
- The panel is the sentence that makes the correction stick — the exponent counts appearances.
- OFF THE SLIDE, YOURS TO SAY: The small 2 says how many times the number appears, not what to multiply it by.

Slide 4. Notes: II. The perfect squares, to 225 3 min

- Read the list aloud as a list, and come back to it every day this unit.
- The panel is a real boundary from the benchmark, not a guess — perfect squares up to 225 and no further, so a student who knows fifteen numbers has the whole range.
- Have them write the list somewhere they will find it again; it is the single highest-value thing on today's page.
- OFF THE SLIDE, YOURS TO SAY: 225 is the largest perfect square the Grade 8 FAST will ask you about.

Slide 5. Notes: III. A square's area runs both ways 3 min

- Item C is where the class slows down.
- Model 'what times itself gives 36?' out loud rather than reaching for a calculator — the list from Notes II is the tool.
- The panel's last clause is a promise, not a rule to memorize: today every answer is positive because every answer is a length.
- Tomorrow the question becomes $x^2 = 36$ rather than 'find the side', and the negative solution appears.
- Flagging it now stops it feeling like a contradiction on day two.
- OFF THE SLIDE, YOURS TO SAY: Area is in square units. A side is a length, so it is in units — and it is positive.

Slide 6. Notes: IV. Squaring never gives a negative answer 3 min

- THE NEW IDEA IN THIS LESSON, and the one the whiteboard sequence tests.
- Item B is the second misconception on Florida's list: students forget the parentheses and compute -3^2 as -9 .

- Write both on the board, one above the other: $(-3)^2 = 9$ and $-3^2 = -9$.
- They are different expressions.
- The panel corrects Math Nation's own page, which says the graph 'only has positive y-values' — false at the origin, and the worksheet asks about it, so say the precise version now.
- OFF THE SLIDE, YOURS TO SAY: “Never below the axis” is correct. “Always above it” is not.

Slide 7. Example 1 *1 of the 6 min*

- SPOKEN ONLY.
- No pencils.
- The answer to the third column is the same three times — multiply the number by itself — and hearing that said aloud before any arithmetic is the point of the slide.
- What changes is only what kind of number you are multiplying.
- Expect someone to say the fraction is different in kind; it is not, and the next slide shows why.

Slide 8. Example 1 *1 of the 6 min*

- They compute all three on whiteboards before anything is revealed.
- Holding the reveal until every board carries three answers is deliberate: a student who does the easy one and waits for the other two is exactly the student this slide is for.
- OFF THE SLIDE, YOURS TO SAY: Write all three. Do not put your board up until you have all three.

Slide 9. Example 1 *1 of the 6 min*

- ANSWERS: 144 square units; $4/25$; 0.49 square units. The fraction row is the one to slow down on — $(2/5)^2$ is $4/25$, not $4/5$ and not $2/25$. The decimal row catches the student who answers 4.9, which is 0.7 multiplied by 7. Both errors are the same error: operating on one part of the number instead of the whole of it.
- OFF THE SLIDE, YOURS TO SAY: Square the top AND the bottom. Two decimal places in, two decimal places out.

Slide 10. Example 1 *1 of the 6 min*

- Nothing on the slide but the question.
- The number is deliberately not a whole number, so the list from Notes II does not answer it and guess-and-check has to become systematic.
- Give them thirty seconds before the next slide.

Slide 11. Example 1 *1 of the 6 min*

- This is the method, not the answer.
- Bounding first and refining second is the same move students will use in Lesson 4 on irrational numbers, so name it now: find the two whole numbers it sits between, then close the gap.
- Do not reveal 1.5 until somebody has proposed it and squared it.
- OFF THE SLIDE, YOURS TO SAY: So the side is between 1 and 2. Two whole numbers, and the answer is neither.

Slide 12. Example 1 *1 of the 6 min*

- Check it in front of them: 1.5×1.5 , worked as $15 \times 15 = 225$ with the decimal point replaced.
- The panel is the honest flag — this one comes out exactly and most do not, which is the whole reason Lesson 4 exists.
- Do not oversell it as a technique.
- OFF THE SLIDE, YOURS TO SAY: Halfway between 1 and 2, and it lands exactly. That will not always happen — Lesson 4 is where it stops happening.

Slide 13. Example 2 *2 of the 5 min*

- Whiteboards, all seven, before anything is revealed.
- The negatives are the reason this table exists — a student who leaves the top three blank has not read Notes IV, and you will see that from the front of the room in one glance.

Slide 14. Example 2 *1 of the 5 min*

- Point at the pairs as you check them: -3 and 3 both give 9 , -2 and 2 both give 4 .
- That is Notes IV item B doing work, not a coincidence about pictures.
- The 0 row is the one to pause on — it is the only value of x that gives 0 , and it is what makes 'never below the axis' the right phrasing.
- OFF THE SLIDE, YOURS TO SAY: Read the column downwards: $9, 4, 1, 0, 1, 4, 9$. It comes back up.

Slide 15. Example 2 *2 of the 5 min*

- Plot the seven points from the table onto the board copy yourself so the class watches it being built rather than seeing it arrive finished.
- Do NOT name it a parabola and do not call it a quadratic — neither is needed for this benchmark and both invite questions the unit cannot answer yet.
- The one point to slow down on is $(0, 0)$: it is ON the axis, which is exactly the distinction from Notes IV.
- OFF THE SLIDE, YOURS TO SAY: Different x , the same y — and the mirror line is the y -axis.

Slide 16. Example 3 *1 of the 5 min*

- Take a show of hands before any working.
- Most classes split.
- Do not settle it — the next slide settles it, and the disagreement is what makes them watch.

Slide 17. Example 3 *2 of the 5 min*

- Let the class establish this themselves; it takes thirty seconds.
- Every whole number is either too small or too big, so the side is not a whole number.
- That much is certain and it is as far as arithmetic takes them today.
- Stop here and let the gap sit before the next slide.

Slide 18. Example 3 *2 of the 5 min*

- THE SENTENCE THAT MATTERS.
- Do NOT let the class settle on 'it must be a decimal or a fraction' — that is what Math Nation's printed answer says and it is false.

- You do not have to use the word irrational today, but do not teach the opposite of it.
- Worksheet question 11 asks exactly this, so what you say here is what you will read back tonight.
- OFF THE SLIDE, YOURS TO SAY: So the answer is YES, the square exists — its side is just not a number you can write down exactly. That is Lesson 4, and it is where irrational numbers come from.

Slide 19. Whiteboards · Question 1 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 20. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline. $6 \times 6 = 36$.
- Everything after this changes exactly one thing about this question, so a wrong answer later names the thing that changed.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 21. Whiteboards · Question 2 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 22. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the base is now negative, inside parentheses.
- $(-6) \times (-6) = 36$.
- SAME ANSWER as question 1, and that is the whole point — a negative times itself is positive.
- Notes IV item B.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 23. Whiteboards · Question 3 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 24. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed from question 2: the parentheses are gone. -6^2 means $-(6 \times 6) = -36$, because the exponent attaches to the 6 only.
- A student who answers 36 here has not seen that the parentheses were doing work.
- This exact distinction is a FAST trap and it costs points every year.

- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 25. Whiteboards · Question 4 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 26. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a decimal base, parentheses back. $0.6 \times 0.6 = 0.36$.
- The board to look for says 1.2 — that is 0.6 doubled, which is Notes I item C surviving four slides.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 27. Whiteboards · Question 5 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 28. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a fraction base.
- Square the top AND the bottom: $4/9$.
- Expect $2/9$ and $4/3$ — each of those squared one part of the fraction and left the other alone.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 29. Whiteboards · Question 6 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 30. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the direction.
- Question 5 gave the side and asked for the area; this gives the area and asks for the side.
- Same two numbers, opposite question.
- A student who answers $16/81$ has squared again instead of undoing it.
- Close by asking why the answer is not also $-2/3$ — because s is a length, which is Notes III, and because tomorrow that answer changes.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 31. Whiteboards · Question 7 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 32. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from question 6: the direction goes back to evaluating, but now TWO expressions have to be evaluated before anything can be compared.
- This is questions 2 and 3 of this round standing side by side, under time pressure, and it is the single most reliable FAST trap in the lesson.
- THE BOARD TO LOOK FOR SAYS 81 AND 81 — that student read $(-9)^2$ correctly and then read -9^2 as though the parentheses were still there.
- A board saying -81 and -81 is the same error mirrored.
- Neither is a slip; both are the parentheses going unread, and the fix is to say aloud what the exponent is attached to.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 33. Whiteboards · Question 8 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 34. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the FAST select-all format, which Math Nation never uses and which this class will meet on the test.
- Three are perfect squares and three are not.
- THE THREE BOARDS TO LOOK FOR, each a different named error: a board with 8 ticked has taken 8 for a perfect square because $2 \times 2 \times 2 = 8$ — that is a CUBE, and it is Lesson 3, so name it as tomorrow rather than as wrong; a board that leaves 1 out has not counted $1 \times 1 = 1$, which is the omission Notes II's list is built to prevent; a board with 20 or 45 ticked has spotted that each is a square times something (4×5 and 9×5) and stopped one step early — that is genuinely Lesson 4's idea arriving three days sound.
- Take all three answers aloud.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 35. Whiteboards · Question 9 of 9 — ask *about 15 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 36. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.

- Croix on his quizzes: "I'm gunna say yes to written work, but I'm probably not going to grade super strictly about written work." With the worksheet gone this is the only place in the period that habit gets built — IXL takes an answer and never sees the work.
- ANSWER: yes, and the side is between 2 and 3, because $2 \times 2 = 4$ is too small and $3 \times 3 = 9$ is too big, and 8 sits between 4 and 9.
- The answer that is not enough is "yes, 2.8" with nothing else: it is close to right, it came off a calculator, and it does not name the two products that prove it.
- What earns the mark is the two products.
- This is Example 3 again, and it is where Lesson 4 comes from — the exact side is not a number you can write down.
- COLOR: none on this sequence. One expression and its value is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 37. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.02.01 — Answer Key

1. the table	1, 4, 9, 16, 25, 36, 49, 64, 81, 100 A student who writes 20 under $s = 10$ has multiplied by 2 instead of squaring — the misconception Florida names first for this benchmark.
2. select all	1, 16, 36, 49, 64, 100 Six of the ten. 8, 20, 45 and 60 are the distractors. Ask which whole number was squared; a student who cannot name it has not shown the answer.
3. side 12	144 square units 12×12 .
4. side $\frac{2}{5}$	Area $\frac{4}{25} \text{ cm}^2$ Square the numerator and square the denominator. Not $\frac{4}{5}$.
5. side 0.7	0.49 square units Two decimal places in, two out. A student who answers 4.9 multiplied by 7.
6. area 81	9 inches
7. area $\frac{4}{25}$	Side $\frac{2}{5} \text{ cm}$ $2 \times 2 = 4$ and $5 \times 5 = 25$.
8. area 2.25	1.5 units Only the positive value is used — a side is a distance. The negative one arrives in Lesson 2.
9. the table and graph	(-3, 9) (-2, 4) (-1, 1) (0, 0) (1, 1) (2, 4) (3, 9) Seven points, all on gridlines. Opposite x-values give the same y-value.
10. below the axis?	No — and it touches the axis at (0, 0). The point needs both halves: why y is never negative AND what happens at $x = 0$. “The graph is always positive” has missed the origin.
11. area of 8	Between 2 and 3, and not a whole number. $2 \times 2 = 4$ is too small and $3 \times 3 = 9$ is too big. Do NOT accept “the side is a decimal or a fraction” — it is neither, though students are not expected to say so until Lesson 4.

Question 11 is the first of the two to circulate for; question 2 is the other, below. It is slides 16–18 in writing. The answer that earns the point names 2 and 3 and gives both products as the reason. The answer to watch for and correct is “the side is a decimal or a fraction” — that is what Math Nation’s own key says, and it is false. The side of a square with area 8 is irrational.

Question 2 is the format to talk about. Select-all appears on the FAST and nowhere in the Math Nation materials. Students who tick the three obvious squares and stop have not checked the rest of the list. Ask for the squared number beside every tick.

Additional Practice 7A.02.01 — Answer Key

1. the table	36, 49, 64, 81, 100, 121, 144, 169, 196, 225 225 is the ceiling of the benchmark. A student who writes 30 under $s = 15$ has multiplied by 2.
2. select all	4, 25, 81, 121, 169 Five of the ten: 2, 5, 9, 11 and 13 squared.
3. side 15	225 square units
4. side $\frac{3}{4}$	Area $\frac{9}{16} m^2$ Square the top and square the bottom. Not $\frac{6}{8}$.
5. side 1.2	1.44 square units A student who answers 2.4 has doubled.
6. area 169	13 inches
7. area $\frac{9}{16}$	Side $\frac{3}{4} m$
8. area 0.64	0.8 units 0.08 squared is 0.0064, which is far too small.
9. the table and graph	(-3, 9) (-2, 4) (-1, 1) (0, 0) (1, 1) (2, 4) (3, 9) Point at (-3, 9) and (3, 9): different x, same y.
10. the classmate	Not quite. It never goes below, but at $x = 0$ it touches. Same distinction as worksheet question 10, asked from the other side. “Never below” is correct; “always above” is not.
11. area of 20	Between 4 and 5, and not a whole number. $4 \times 4 = 16$ too small, $5 \times 5 = 25$ too big.

This sheet is optional. Post it to Google Classroom for students who want more. It is not assigned and it is not collected.

Changes from the Math Nation Lesson

- Every partner and group activity is replaced — eight printed instances plus the Stronger and Clearer Each Time routine. The two Explorations are now Notes II and III with Examples 1 and 2; the definition-collecting table with initials in Bring It Together is now a vocabulary band in Notes I.
- **A false sentence was corrected.** Math Nation's 2.1.4 completes the statement “for any value of x multiplied by itself, the value of y will always be positive,” then adds “so all of the points other than $(0,0)$ are above the x -axis.” The sentence contradicts its own ending: at $x = 0$ the value of y is 0, which is not positive. The notes, the worksheet and the Additional Practice all now say “never below the axis, and it touches the axis at $(0, 0)$.”
- **A printed answer that teaches a misconception was rewritten.** Math Nation's answer to “can a square have an area of 8?” reads “it is possible if the measure of the sides of the square are decimals or fractions.” The side is $\sqrt{8} \approx 2.828\dots$, which is irrational — neither a fraction nor a terminating decimal. Since MA.8.NSO.1.1 is assessed on exactly this distinction, the key now says “between 2 and 3, and not a whole number,” and the teaching note says explicitly not to accept the decimal-or-fraction answer.
- The warm-up was replaced. Math Nation's is a growth-mindset reflection with two “answers will vary” questions and no numbers in it. The same-and-different on 7^2 and $(-5)^2$ takes the same four minutes and puts the lesson's whole question on the board before a word is said.
- The wrap-up was replaced. Math Nation's asks students to plot a dot on an interesting-versus-confusing grid and write a question they still have. The Ticket Out the Door produces three answers you can read in thirty seconds, and its third question predicts who is ready for Lesson 2.
- A select-all item was added, because that is a FAST format and it does not appear anywhere in the Math Nation unit.
- “Building On: N/A” was filled in. Students have met exponents in MA.6.NSO.3.3 and MA.7.NSO.1.1 and area in grade 6; what is new is reversing the operation.
- Answer keys were built for both sheets, from the same source file as the student versions, so the two cannot drift apart.
- **Nothing in the mathematics was changed** apart from the two corrections above, both of which were printed statements that are false.